Bijal Panchal

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SUMMARY

Data Analyst with a Master of Data Analytics (Distinction) and 3 years of experience in star-schema warehousing, automated pipelines, and predictive modeling. Cut data inconsistencies by 30%, improved query performance by 40%, and built a dropout-risk model with 0.89 AUC-ROC.

SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, Statsmodels, SciPy, Matplotlib, Seaborn), SQL (BigQuery SQL, PostgreSQL), VBA, SQLAlchemy
Modeling & Analysis: K-Means Clustering, XGBoost, PCA, ARIMA, SARIMA, Holt-Winters, Prophet, Isolation Forest, Feature Engineering, Model Interpretability, Model Evaluation & Validation (Precision, Recall, F1-Score, AUC-ROC), Train-Test Validation, A/B Testing, Hypothesis Testing, Confidence Intervals & Effect-Size Estimation
Data Quality & Documentation: Standardized Workflow & Process Development, Data Quality Management, Automated Data Testing, Documentation & Traceability, Root Cause Analysis, Reconciliation
BI/Visualization: Tableau, Power BI (DAX, Power Query), Streamlit, Excel, PowerPoint, Word
Data & Warehousing: Structured & Unstructured Data, MySQL, SQLite, PostgreSQL, Star Schema Design, ETL/ELT Pipelines, Dimensional Modelling, Analytics Engineering (dbt models: staging and marts), API Integration
DevTools: Git, GitHub, GitHub Actions (CI/CD), Jupyter Notebook, VS Code, pytest
CloudStack (developing): GCP BigQuery, dbt Core, Azure, Databricks, Snowflake, AWS S3/Redshift

WORK EXPERIENCE

POINTS AFRICA Aug 2025 - Feb 2026
Data Analyst Intern (Remote) Toronto, ON

- **Architected a PostgreSQL star-schema data warehouse** unifying data from 3 enterprise partners (GLICO, JUMIA, STAR) using Python and SQL, **reducing data inconsistencies by 30%** and **improving query performance by 40%**.
- **Cleaned and standardized transactional, loyalty, and geographic datasets at scale**, resolving formatting errors and duplicates to achieve **100% field coverage** across all partner sources.
- **Applied RFM segmentation** to classify customer loyalty tiers, identifying high-value clusters across all 3 partner networks and **informing targeted engagement strategies**.
- Delivered **10+ Tableau dashboards** tracking KPIs, partner performance, and regional trends, **adopted by senior stakeholders** to drive decisions.

DMG GROUP OF COMPANIES Jan 2023 - Jul 2025
Data Analyst (Remote) Ahmedabad, India

- Engineered a PostgreSQL star-schema warehouse consolidating enrollment, LMS, and assessment data for **100,000+ learners**; wrote advanced SQL for extraction, transformation, and validation, creating a **single source of truth** for academic reporting.
- Developed an XGBoost dropout-risk model with **0.89 AUC-ROC** and **0.82 F1-score**, allowing academic teams to **flag at-risk students early** and prioritize intervention before dropout.
- Created Power BI dashboards and Excel reports on high-risk cohorts, enrollment trends, and model performance, giving non-technical staff **direct access to insights** without writing queries.
- Launched a real-time Streamlit early-warning dashboard and **automated model monitoring** (PSI/KS-test drift detection) to maintain prediction accuracy as new data flowed in.

PROJECTS

City Traffic Intelligence Platform | Python, SQLite, Streamlit, Scikit-learn, Prophet | Live

- Engineered a **fully automated real-time data platform** ingesting daily metrics across **5 global cities** from **5 live APIs** (TomTom, OpenAQ, Open-Meteo, Numbeo, City Open Data) — built on modular Python collectors feeding a SQLite star schema warehouse (**9 tables, 4,400+ rows**) via a self-running GitHub Actions pipeline.
- Designed a **weighted City Stress Index scoring engine** (Traffic 35% + Air Quality 25% + Weather 20% + Safety 10% + Cost of Living 10%) powered by **15+ engineered features** — including rolling averages, volatility scores, and heat stress index — using ARIMA, Prophet, Z-Score, Isolation Forest, and K-Means models.
- Deployed a **7-page interactive Streamlit dashboard** (live at city-stress-index.streamlit.app) with real-time stress scores, 30-day forecasts, anomaly detection alerts, and cross-city clustering — validated by a **22-test pytest suite** ensuring pipeline and database integrity.

Retail Sales Forecasting & Demand Planning | Python, SQL, Power BI, Statsmodels, SQLite

- Cleaned and processed **1M+ transactional records** across **43 countries and 4,735 products**, resolving 34K duplicates, 19K cancellations, and 243K missing customer IDs — then engineered a SQL star schema warehouse (1 fact + 4 dimension tables) via a custom Python ETL pipeline (SQLAlchemy).
- Built and benchmarked **SARIMA and Holt-Winters** forecasting models on monthly revenue data, with SARIMA achieving **74.5% accuracy** (MAPE 25.54%) on a chronological 80/20 train-test split — reducing simulated stockout rates by **18%**.
- Delivered a **4-page Power BI dashboard** with dynamic KPI cards, seasonal trend lines, and forecast-vs-actual comparisons (80% confidence intervals) — powering RFM segmentation across 6 customer tiers that identified **843 Champions** (avg. $5,689 spend) and flagged 409 At-Risk customers as the top re-engagement priority.

ExperimentFlow: Cloud Analytics & A/B Testing Platform | In Progress
Python, SQL, Google BigQuery, dbt Core, GitHub Actions, SciPy, Power BI

- Developing a **cloud-based analytics pipeline** to ingest and transform **100K+ e-commerce orders** and **50K+ simulated website experiment events** using **Python, SQL, BigQuery, and dbt**, organized into **15+ staging and mart models** on a **star schema**.
- Implementing **25+ automated dbt data-quality tests** and **scheduled GitHub Actions CI workflows** (on every pull request and daily) to improve **data reliability and reproducibility**.
- Conducting **A/B test analysis** across **3 experiments** using **conversion rates, 95% confidence intervals, hypothesis testing, and effect-size estimates** to support **evidence-based product decisions**.
- Building a **4-page Power BI dashboard** to communicate **retail performance, experiment outcomes, and pipeline health** to **business stakeholders**.

EDUCATION

University of Niagara Falls Oct 2024 - May 2026
Master of Data Analytics (MDA)

- Graduated with Distinction | GPA: 3.89/4.00 | President's Honour Roll

Navrachana University Jul 2020 - Jun 2024
Computer Science & Engineering (B.Tech)

CERTIFICATION

Advanced Power BI: DAX Language, Formulas, and Calculations | Course Completion – Apr 2025

- A targeted Power BI certification covering core skills in data modeling, DAX language, and advanced DAX formulas. Emphasizes practical application through dynamic dashboards and real-world business intelligence scenarios.